CSI-Sense: Detecting Presence Behind Obstructions For S.A.R. Using WiFi

Final Report for Computer Systems Lab

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Abstract

Search-and-rescue operations after structural collapse require responders to locate trapped individuals quickly while avoiding unnecessary entry into unstable environments. Traditional sensing tools such as cameras, thermal imaging, acoustic detection, and commercial radar either require line of sight, depend on victim movement or sound, or rely on expensive specialized hardware. This project, CSI-Sense, explores an alternative approach using commodity Wi-Fi hardware and Channel State Information (CSI) to detect human presence behind obstructions. A 2.4 GHz Wi-Fi router transmits packets while a Raspberry Pi 4 Model B running Nexmon CSI captures packet-level CSI measurements. The resulting PCAP recordings are converted into CSI matrices, cleaned, segmented into time windows, and processed into amplitude-based frequency features using a Short-Time Fourier Transform (STFT). A binary Random Forest classifier then predicts whether the monitored region is vacant or occupied. The system achieved 90.0% file-level accuracy on a held-out test split, including 80.0% vacant accuracy and 93.3% occupied accuracy. A red intensity overlay was also created to visualize model confidence and motion-related CSI energy over time. Testing showed that the system can detect presence behind a door, although cluttered and uncontrolled environments remain challenging. Overall, CSI-Sense demonstrates that low-cost Wi-Fi CSI can serve as a practical prototype for non-line-of-sight presence detection, while also identifying key limitations that must be addressed before real deployment in SAR environments.

I. Introduction

Search and rescue (SAR) operations in post-disaster environments present a fundamental challenge: locating survivors quickly enough in conditions where direct line of sight is unavailable and physical access is dangerous. After earthquakes, explosions, or structural failures, buildings may become unstable debris fields with hidden voids that could contain trapped survivors. Entering every possible void is not realistic because responders may be injured or may worsen the instability of the structure. As a result, search efforts are often slow, selective, and hazardous.

Time is the most critical factor. The survival window is commonly discussed as the first 24 to 72 hours after entrapment, meaning that delays in locating victims can significantly reduce the chance of survival. Traditional methods struggle in these environments. Cameras and visual inspection fail when debris, walls, smoke, dust, or darkness block visibility. Acoustic methods depend on victims being conscious and able to make sound. Thermal cameras detect surface heat but cannot reliably see through thick barriers or dense debris. As a result, responders often operate with incomplete information.

This project addresses the problem of detecting human presence behind obstructions without requiring direct line of sight. Instead of attempting the more complex task of estimating body pose or identifying detailed activity, the project narrows the objective to binary presence detection: determining whether a hidden monitored region is likely vacant or occupied. This simplified objective is better matched to SAR needs, because even a binary signal can help responders prioritize where to search first.

The core idea is that Wi-Fi signals interact with the surrounding environment. They pass through some materials, reflect off surfaces, and scatter off objects including the human body. A person standing, breathing, or moving behind an obstruction changes the wireless channel in small but measurable ways. By analyzing these changes through Channel State Information (CSI), CSI-Sense attempts to infer whether human presence exists behind a barrier.

II. Background

Wi-Fi is a radio frequency (RF) communication technology that transmits data through electromagnetic waves, typically in the 2.4 GHz and 5 GHz bands. In this project, 2.4 GHz Wi-Fi was used because lower-frequency Wi-Fi generally has better range and wall penetration than 5 GHz Wi-Fi. In an indoor environment, Wi-Fi signals do not travel only in a straight line. They reflect, scatter, and combine through multipath propagation, producing a channel pattern that changes when a person or object moves.

Two common measurements can be extracted from Wi-Fi behavior: Received Signal Strength Indicator (RSSI) and Channel State Information (CSI). RSSI gives a coarse single value representing received signal power. CSI is more detailed because it describes how each OFDM subcarrier is affected by the environment. Since Wi-Fi uses Orthogonal Frequency Division Multiplexing (OFDM), a wideband signal is divided into many subcarriers, and CSI provides amplitude and phase information for those subcarriers.

A CSI value can be represented as a complex number:

```
H = a +  b
|H| = sqrt ( a^2 + b^2)                    #amplitude
Theta = atan2(b, a)                         # phase
```

The amplitude describes signal strength, while phase describes relative timing or shift. Over time, changes in amplitude and phase can correspond to motion, breathing, or environmental disturbance. This project primarily uses amplitude-derived features because they were easier to stabilize for the available data and time constraints.

To capture CSI, this project uses Nexmon CSI, an open-source framework that exposes CSI from compatible Broadcom Wi-Fi chips. In this setup, a Raspberry Pi 4 Model B captures CSI and stores packet captures as PCAP files. These PCAP files are later converted into Python pickle files containing CSI matrices and timestamps.

III. Related Work

Prior RF sensing work has shown that radio signals can be used for non-line-of-sight perception. RF-Pose, developed by researchers at MIT, uses deep neural networks to estimate 2D human skeletal pose through occlusions and walls. The system demonstrates that RF reflections can encode human movement even when cameras cannot see the subject [1]. However, RF-Pose requires substantial synchronized RF and vision data and complex deep learning methods, which makes it difficult to reproduce in a small student-built prototype.

Other Wi-Fi CSI systems focus on activity recognition, such as classifying walking, sitting, or other indoor activities. These systems show that CSI contains useful motion-related patterns, but they usually require large labeled datasets and controlled environments. Through-wall radar systems, especially ultra-wideband (UWB) radar, can detect motion and breathing behind barriers, but commercial SAR-grade radar systems are expensive and less accessible to small-scale responders.

CSI-Sense differs from these systems by intentionally reducing the problem scope. Instead of performing pose estimation or multi-class activity recognition, it performs binary vacant-versus-occupied classification. This makes the project more feasible with limited data, limited computation, and commodity hardware while still addressing a practical SAR question: is there likely a person behind this obstruction?

IV. Novelty and Advantages

The novelty of CSI-Sense is its low-cost, binary presence-detection approach using commodity Wi-Fi hardware. Many existing RF-based sensing systems require specialized radar hardware, antenna arrays, or deep neural networks trained on large datasets. CSI-Sense instead uses a Raspberry Pi, a standard 2.4 GHz Wi-Fi router, and open-source CSI extraction tools. This lowers the barrier to entry and makes the concept more accessible for small teams, volunteer groups, school labs, and community responders.

The project is also novel in its SAR framing. In a disaster response setting, responders do not always need a full pose estimate or detailed activity label. A reliable indication that a hidden region is likely occupied can already improve decision-making. By focusing on presence detection, the model can remain simpler and more interpretable than a neural-network pose estimation system.

Another advantage is scalability. Because the hardware is inexpensive compared with commercial UWB radar, multiple sensing units could eventually be deployed across a wider disaster area. A large number of low-cost nodes could help responders identify promising search areas faster, especially in environments where line-of-sight tools fail.

V. Impact

The main impact of this project is the possibility of making non-line-of-sight sensing more accessible. A tool like CSI-Sense could eventually help small-scale responders, NGOs, local emergency teams, and firefighters perform preliminary scans without relying on expensive radar systems. This does not replace trained SAR procedures, but it could provide an additional layer of situational awareness.

CSI-Sense also has safety implications. If responders can scan a questionable area from a distance, they may avoid entering unstable structures unnecessarily. Even a rough detection signal could help teams prioritize where to investigate first, reducing wasted effort in low-probability zones.

Finally, the system supports scalable SAR operations. In large structural failures, a single expensive sensor may not cover enough area. A network of low-cost Wi-Fi CSI nodes could potentially provide broader coverage, especially in settings with limited funding or limited access to specialized equipment.

VI. Methods

A. System Overview

The CSI-Sense system uses a 2.4 GHz Wi-Fi router as the transmitter and a Raspberry Pi 4 Model B running Nexmon CSI firmware as the receiver. The router transmits Wi-Fi packets while the Raspberry Pi records CSI measurements from the received packets. These measurements describe how the wireless channel changes across subcarriers over time. The data is then moved to a laptop for preprocessing, feature extraction, training, classification, and visualization.

Fig. 1: CSI-Sense System Architecture

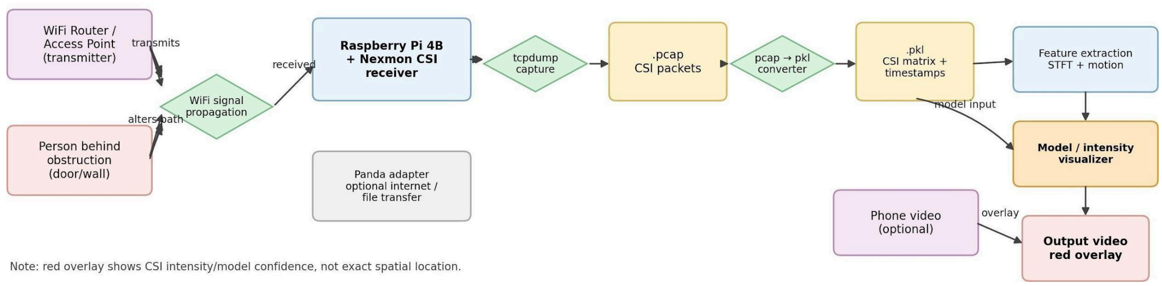


Fig. 1. System architecture for CSI-Sense, showing Wi-Fi transmission, CSI capture, PCAP conversion, feature extraction, classification, and red-overlay visualization.

B. Materials and Tools

The hardware and software materials are listed in Table 1. The Raspberry Pi served as the CSI receiver, the Wi-Fi router served as the transmitter, and the laptop handled preprocessing, modeling, and visualization.

Category	Item	Purpose
Hardware	Raspberry Pi 4 Model B	Receiver used to collect CSI using Nexmon CSI
Hardware	2.4 GHz Wi-Fi router	Transmitter that sends packets through the monitored environment
Hardware	Laptop	Preprocessing, model training, graphing, and video generation
Software	Nexmon CSI	Exposes CSI from compatible Wi-Fi chips
Software	tcpdump	Captures CSI packets into PCAP files
Software	CSIKit	Parses CSI data from PCAP files
Software	Python, NumPy, SciPy	Numerical processing and STFT feature extraction
Software	scikit-learn	Random Forest classifier and evaluation metrics
Software	OpenCV, Matplotlib, joblib	Video generation, graphing, and model persistence

C. Input and Output

The raw input is a PCAP file containing captured CSI packets. After conversion, each recording becomes a PKL file containing a CSI matrix and timestamp array. The expected format is a matrix with subcarriers along one dimension and time samples along the other. The model output is a binary prediction: vacant or occupied. The visualization output is an MP4 video with a red intensity overlay that changes based on the model probability and motion-related CSI energy.

Stage	Input	Output
Capture	Wi-Fi packets received by Raspberry Pi	PCAP file
Conversion	PCAP file	PKL file with valid_csi and timestamps
Feature extraction	CSI matrix	Low, mid, high, and entropy features per window
Classification	Feature matrix	Vacant or occupied prediction
Visualization	Predictions and motion energy	Red intensity overlay video

D. Data Collection

Data collection was performed using a controlled indoor setup. The router and Raspberry Pi remained in fixed positions while the monitored region behind the obstruction changed. The obstruction used in the final setup was drywall. Three complexity conditions were defined: simple, moderate, and complex. The simple condition used only a single drywall barrier. The moderate condition added light clutter such as a chair or box. The complex condition added heavier clutter such as stacked boxes, a table, bins, or other objects meant to imitate a more obstructed space.

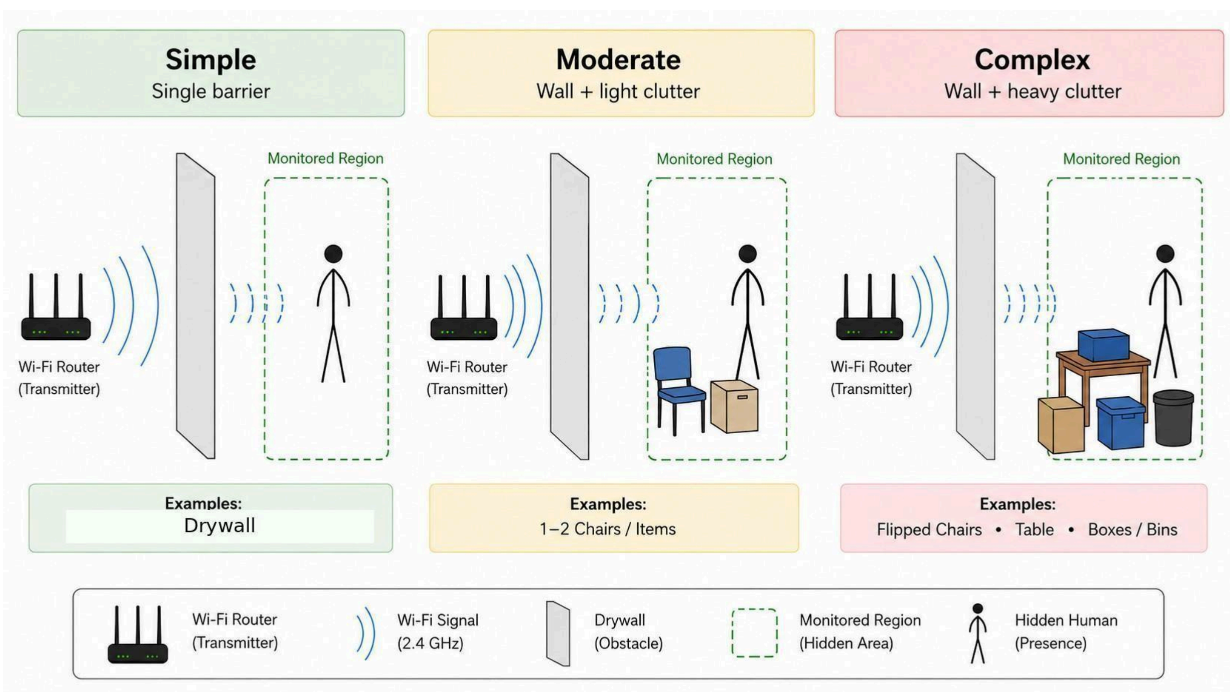


Fig. 2. Monitored-region complexity levels used for CSI-Sense testing. The final setup used drywall as the barrier, with complexity increasing through added clutter.

For each condition, a baseline recording with no person present was compared against human-present recordings. The human-present recordings included standing, stable presence, or movement behind the obstruction. The Raspberry Pi recorded packet-level CSI as PCAP files. The recordings were later transferred to the laptop and converted into the model input format.

E. Preprocessing

The preprocessing stage converts PCAP files into PKL files. During conversion, each CSI frame is extracted, squeezed into a consistent array shape, and checked for invalid values. Frames containing non-finite values are discarded. If the CSI matrix is loaded in the reverse orientation, it is transposed so that the model consistently treats one dimension as subcarriers and the other as time samples.

Some subcarriers can be noisy because of hardware artifacts or unstable capture behavior. The preprocessing and visualization steps therefore focus on stable amplitude patterns and remove obviously corrupted values. Early diagnostic heatmaps were used to identify subcarrier behavior and to compare human-present conditions against baseline conditions.

F. Feature Extraction

The feature extraction stage converts raw complex CSI into a compact numerical representation. First, the complex CSI matrix is converted into amplitude using $|H|$. Then, for each time step, amplitudes are averaged across subcarriers. This produces a one-dimensional signal $sig[t]$ representing how the overall channel strength changes over time.

A Short-Time Fourier Transform (STFT) is applied to each window. STFT converts the time-domain signal into a frequency-domain representation, making it possible to measure how much power exists in slow, medium, and fast motion bands. Four features are extracted from each window:

- Low-frequency energy fraction: power at frequencies <= 0.5 Hz divided by total power.
- Mid-frequency energy fraction: power from 0.5 Hz to 2.5 Hz divided by total power.
- High-frequency energy fraction: power above 2.5 Hz up to 10 Hz divided by total power.
- Spectral entropy: how spread out the power distribution is across frequency bins.

These features were chosen because they are interpretable. Low-frequency energy captures near-static channel behavior, mid-frequency energy captures slower movement and possible breathing-like variation, high-frequency energy captures faster movement, and entropy describes whether motion energy is concentrated or messy.

G. Classification Model

The classifier is a binary Random Forest model. The model predicts whether each CSI window belongs to a vacant or occupied condition. Random Forest was selected because it works well on small tabular feature sets, handles noisy features better than a single decision tree, and is much easier to train than a neural network. No neural network architecture was used in the final system.

Each file contains multiple sliding windows. The classifier produces an occupied probability for each window. The file-level decision is based on the mean occupied probability and the fraction of high-confidence occupied windows. This is more appropriate than reporting only window-level predictions because an SAR system needs to decide whether a whole monitored region is likely occupied.

H. Visualization and Intensity Calculation

The system also generates a red intensity overlay video. This visualization is not a spatial localization map. Instead, it is a confidence-style display that shows when the model and the CSI motion features indicate stronger evidence of presence. The intensity calculation combines model confidence with normalized motion energy:

intensity = p_occ*(motion_floor + 0.75 * motion_norm)

Here, p_occ is the model probability that the current window is occupied, motion_norm is the normalized mid/high-frequency motion energy, and motion_floor prevents visible detections from disappearing completely when model confidence is present. The intensity is smoothed over time to reduce flickering, and a persistence filter removes isolated weak detections.

Fig. 3. Example red intensity overlay. The red blob represents model confidence and motion-related CSI energy, not exact physical location.

I. Estimated Obstruction Complexity

For the final comparison graph, baseline files were treated separately as vacant recordings. Human-present recordings were grouped into simple, moderate, and complex occupied categories based on the intended obstruction/clutter setup and CSI-derived signal distortion. Because the complexity grouping is not a full real-world rubble label, the graph is interpreted as an estimated robustness comparison rather than a complete measurement of physical disaster-scene complexity.

J. Performance Metrics

The main metric is file-level accuracy, calculated as correct file predictions divided by total file predictions. Additional metrics include precision, recall, and F1-score for the vacant and occupied classes. File-level metrics are important because the system is designed to support a decision about whether a monitored region should be considered vacant or occupied.

K. Pseudocode

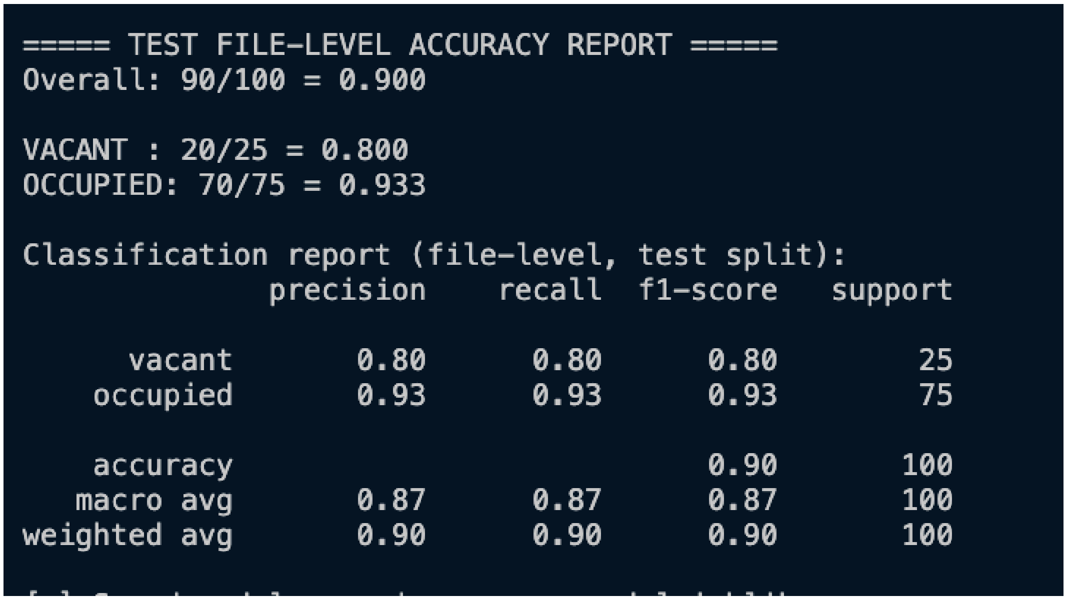
```
procedure COLLECT_CSI
    position router and Raspberry Pi
    place drywall obstruction and optional clutter
    enable Nexmon CSI on Raspberry Pi
    run tcpdump to capture CSI packets
    save recording as PCAP file
end procedure


procedure CONVERT_PCAP_TO_PKL(pcap_file)
    read CSI frames from pcap_file
    remove invalid frames
    reshape frames into consistent CSI matrix
    save valid_csi and timestamps into PKL file
end procedure


procedure EXTRACT_FEATURES(csi, timestamps)
    convert complex CSI to amplitude
    average amplitude across subcarriers
    split signal into 1.5-second windows

    for each window:
        apply STFT
        compute low, mid, and high energy fractions
        compute spectral entropy

    return feature matrix
end procedure


procedure CLASSIFY_FILE(pkl_file)
    load CSI matrix and timestamps
    extract features
    scale features
    compute occupied probability per window
    aggregate probabilities into file-level prediction
    return VACANT or OCCUPIED
end procedure
```

L. What Did Not Work

Several approaches were considered or tested before arriving at the final pipeline. Using only raw RSSI or raw amplitude plots was not reliable because the signal varied with distance, antenna orientation, hardware behavior, and environmental reflections. Deep learning was considered, but it was not practical because it would require far more labeled data and more computation than was available. Early live captures also showed that CSI data can be noisy and that models can become overconfident when the training and live capture environments differ. These issues motivated a lightweight Random Forest pipeline with interpretable features and file-level aggregation.

VII. Results

The final model achieved 90.0% file-level accuracy on the held-out test split. Out of 100 test files, 90 were correctly classified. The vacant class achieved 20/25 correct predictions, or 80.0% accuracy. The occupied class achieved 70/75 correct predictions, or 93.3% accuracy. This indicates that the model was better at recognizing human-present recordings than rejecting all vacant recordings.



```
===== TEST FILE-LEVEL ACCURACY REPORT =====
Overall: 90/100 = 0.900

VACANT : 20/25 = 0.800
OCCUPIED: 70/75 = 0.933

Classification report (file-level, test split):
              precision    recall  f1-score   support

      vacant       0.80      0.80      0.80        25
    occupied       0.93      0.93      0.93        75

    accuracy                           0.90       100
   macro avg       0.87      0.87      0.87       100
weighted avg       0.90      0.90      0.90       100
```

Fig. 4. Terminal output showing the file-level test accuracy report: 90/100 overall, 20/25 vacant, and 70/75 occupied.

Class	Correct / Total	Accuracy	Precision	Recall	F1-score
Vacant	20 / 25	0.800	0.80	0.80	0.80
Occupied	70 / 75	0.933	0.93	0.93	0.93
Overall	90 / 100	0.900	-	-	-

A second analysis compared baseline vacant performance against occupied recordings grouped by estimated obstruction complexity. The baseline vacant group remained at 0.800. The occupied groups averaged approximately 0.933 across simple, moderate, and complex categories. The chart should be read carefully: the complexity groups are based on the experimental setup and CSI distortion, not a complete real-world rubble benchmark.

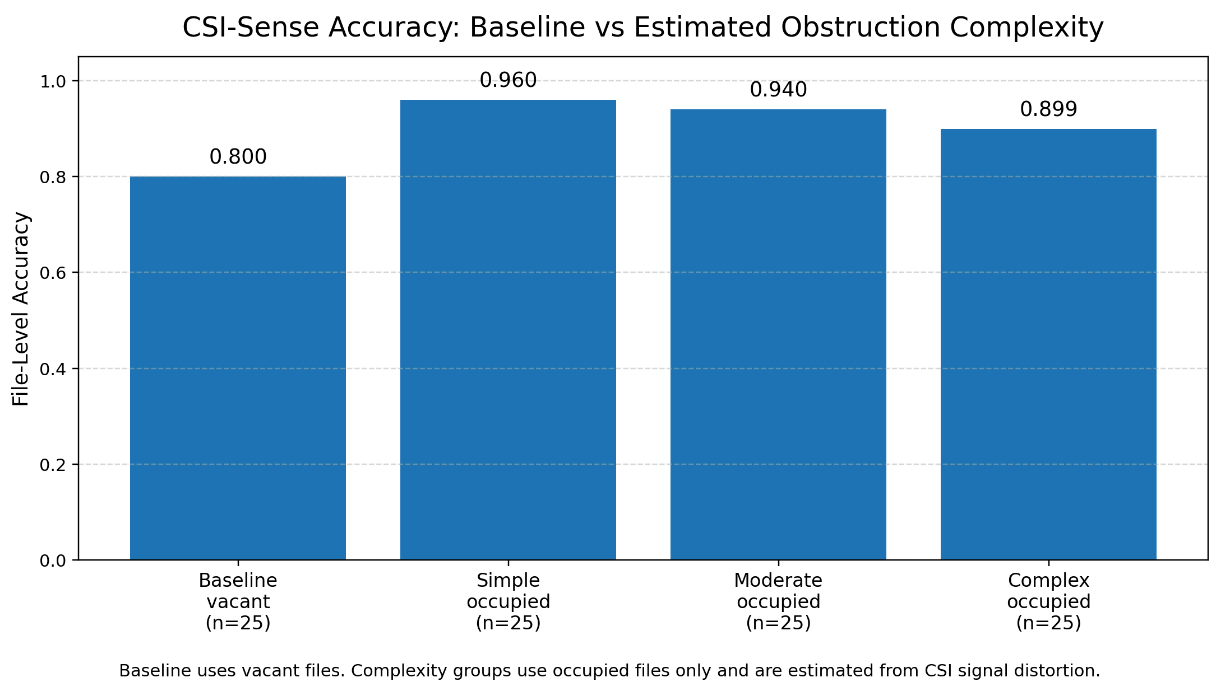


Fig. 5. File-level accuracy by baseline and estimated obstruction complexity. Baseline uses vacant files; complexity groups use occupied files only and are estimated from CSI signal distortion.

The red-overlay demo showed that movement behind an obstruction can cause visible changes in the CSI intensity value. In the one-minute demo, the first half represents a baseline period, and the second half shows movement behind the obstruction. Movement of the body and even leg movement behind a box produced spikes in intensity, supporting the idea that CSI can detect changes in the hidden region.

Early diagnostic heatmap comparisons also showed that human presence can increase CSI variability. Around subcarrier 110 at approximately 2.35 seconds, the sleeping/stable condition showed substantially higher variation than the empty baseline. The measured value was about 31 for the human-present condition compared with about 17 for the empty baseline, suggesting that even quiet human motion or breathing can noticeably affect CSI variability.

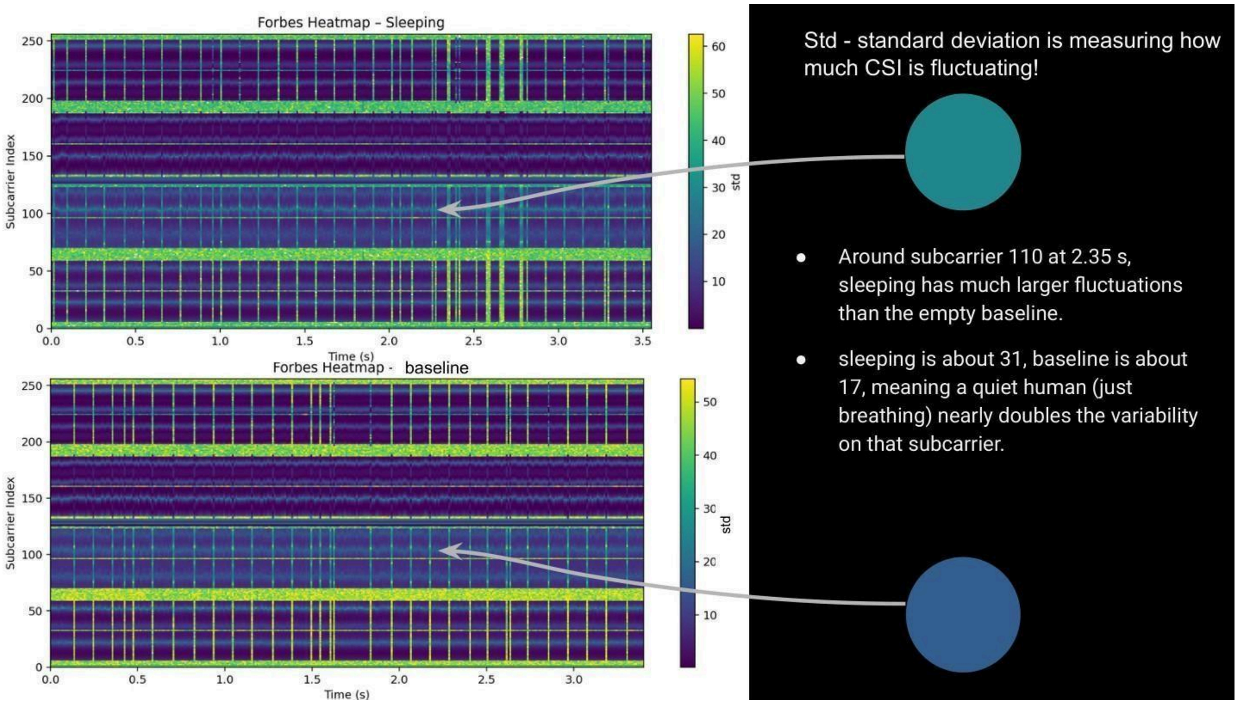


Fig. 6. Heatmap comparison between a stable human-present condition and an empty baseline. Around subcarrier 110 at 2.35 s, the human-present condition shows greater CSI fluctuation.

The most important practical result is that the system successfully detected human presence behind a door in controlled testing. This supports the feasibility of commodity Wi-Fi CSI for low-cost non-line-of-sight sensing. However, results also showed that cluttered environments were more difficult to interpret, and model confidence could become unstable when the environment differed from the training conditions.

VIII. Limitations

CSI-Sense is a prototype and has several important limitations. First, the testing environment cannot fully replicate true SAR conditions. Real disaster sites include unstable rubble, multiple layers of material, dust, smoke, irregular voids, metal objects, moisture, and unpredictable human positions. The project was tested in a controlled indoor environment and therefore cannot claim readiness for actual rescue deployment.

Second, the obstruction conditions were simplified. The final setup used drywall and cluttered objects, but did not systematically test many construction materials, thicknesses, or distances. Radio propagation depends strongly on material type and geometry, so the current results may not generalize to concrete, brick, metal, or dense multi-layer debris.

Third, the system detects presence but does not provide true spatial localization. The red blob visualization is an intensity display, not a map of where the person is located. This is useful for communicating model confidence, but it should not be interpreted as a physical position estimate.

Fourth, the dataset is limited. More recordings across more rooms, people, positions, distances, and clutter arrangements would be needed to improve robustness. The vacant class was also harder for the model than the occupied class, which suggests that false positives remain an important issue.

Finally, the current workflow is not fully real-time in a field-ready sense. CSI is captured on the Raspberry Pi, transferred or processed on a laptop, and then visualized. A deployable system would need more direct device-to-device processing and a more reliable interface for responders.

IX. Conclusion

CSI-Sense demonstrates that commodity Wi-Fi hardware can be used to prototype a non-line-of-sight presence detector for SAR-inspired scenarios. By extracting CSI from Wi-Fi packets, converting it into amplitude-based time-series features, and applying a binary Random Forest classifier, the system achieved 90.0% file-level accuracy on the held-out test split. The model performed especially well on occupied recordings, achieving 93.3% occupied accuracy, and successfully detected presence behind a door in controlled testing.

The project also produced visual outputs that make the detection process easier to interpret. The red intensity overlay shows changes in model confidence and CSI motion energy over time, and heatmap diagnostics show how human presence can alter subcarrier variability. These results support the central idea that Wi-Fi CSI can capture useful information about hidden human presence.

At the same time, the project remains a prototype. It does not yet solve the full SAR problem, and it does not replace professional rescue equipment or procedures. The strongest contribution is demonstrating a low-cost, scalable direction for future exploration: using widely available Wi-Fi devices to provide additional non-line-of-sight sensing information.

X. Future Work and Recommendations

Future work should focus on improving reliability, expanding the dataset, and moving the system closer to real-time deployment. The most important next step is to collect a larger and more systematic dataset. This should include many rooms, multiple people, different distances, and repeated trials across vacant and occupied states. Each recording should be carefully labeled by obstruction type, clutter level, distance, and subject behavior.

A second direction is to systematically study RF penetration through different materials. The current setup primarily used drywall, but real structures may contain concrete, brick, wood, insulation, metal, glass, and combinations of materials. Testing material thickness and distance would help determine where Wi-Fi CSI is useful and where it fails.

Third, the pipeline should be improved for direct real-time use. Instead of capturing on the Raspberry Pi and analyzing later on a laptop, future versions should process CSI directly on the edge device or stream it to a lightweight dashboard. A responder-facing interface should show confidence, warnings, and recent signal history rather than only a side-by-side video.

Fourth, future models should address false positives in vacant conditions. Since vacant accuracy was lower than occupied accuracy, additional baseline data and better negative examples are needed. More robust calibration could help the system distinguish human presence from background Wi-Fi noise, environmental motion, or hardware artifacts.

Finally, the visualization should be improved. The current red blob is useful for communication but does not represent true location. Future work could explore multi-node triangulation, multiple receiver placement, or coarse spatial heatmaps if several Wi-Fi sensing units are deployed together.

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Github Repository

intensity overlay changes. If you look closely, even moving my legs behind the box causes a spike in intensity. This does not show exact location, but it visualizes how low-cost Wi-Fi hardware can reveal motion behind a barrier.

Appendix C: Notes on Interpretation

CSI-Sense should be interpreted as a proof-of-concept prototype. The system shows that commodity Wi-Fi CSI can detect meaningful changes behind an obstruction in controlled tests, but it is not a finished SAR device. Its main value is demonstrating a low-cost sensing direction and identifying the technical work needed for future field-ready systems.